

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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DATE:03/10/2026

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Course Code/ Name: DSA0601 /Data Handling and Visualization

Slot C | CLASS ACTIVITY (R Only)

Chart 1: Fuel efficiency (km per litre)

Violin	Skew (answer)	Mean vs Median (answer)
Brand Z	Right skew	Mean > Median
Brand X	Left skew	Mean < Median
Brand V	Left skew	Mean < Median

Interpretation: Brand Z: Right skew, so Mean > Median.; Brand X: Symmetric, so Mean $\approx$ Median.; Brand V: Left skew, so Mean < Median.

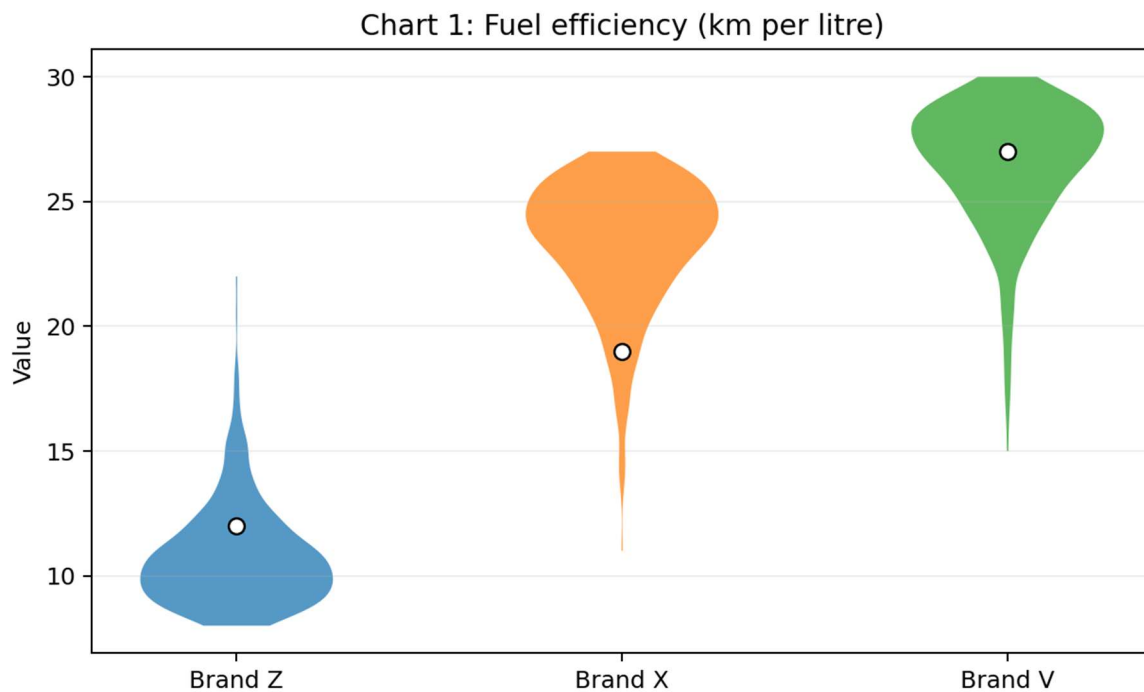


Chart 2: Order value (Rs hundred)

Violin	Skew (answer)	Mean vs Median (answer)
Store 1	Symmetric	Mean $\approx$ Median
Store 4	Left skew	Mean < Median
Store 2	Right skew	Mean > Median

Interpretation: Store 1: Symmetric, so Mean $\approx$ Median.; Store 4: Left skew, so Mean < Median.; Store 2: Right skew, so Mean > Median.

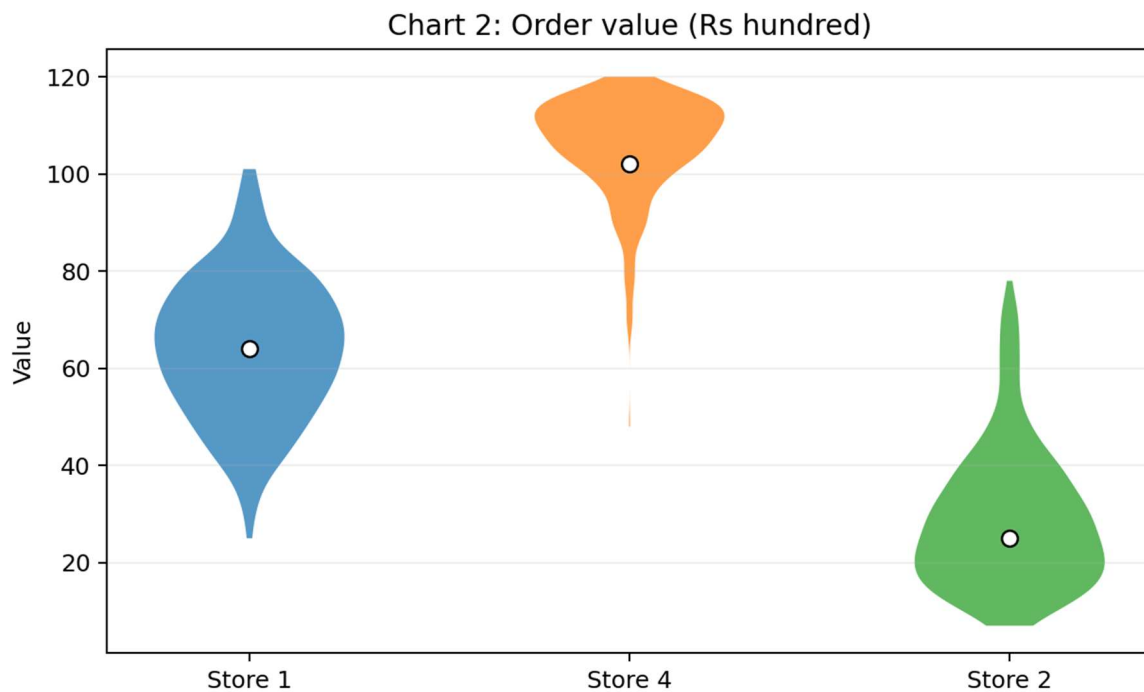


Chart 3: Monthly income (Rs thousand)

Violin	Skew (answer)	Mean vs Median (answer)
City	Left skew	Mean < Median
Metro	Symmetric	Mean $\approx$ Median
Town	Right skew	Mean > Median

Interpretation: City: Left skew, so Mean < Median.; Metro: Symmetric, so Mean $\approx$ Median.; Town: Right skew, so Mean > Median.

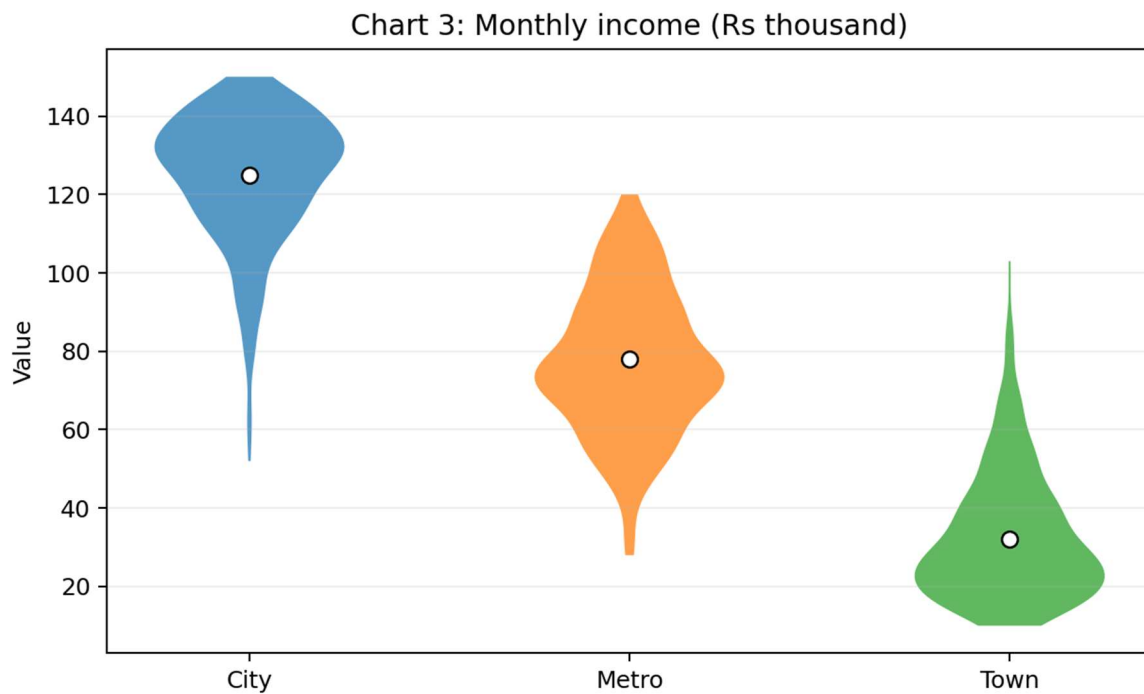
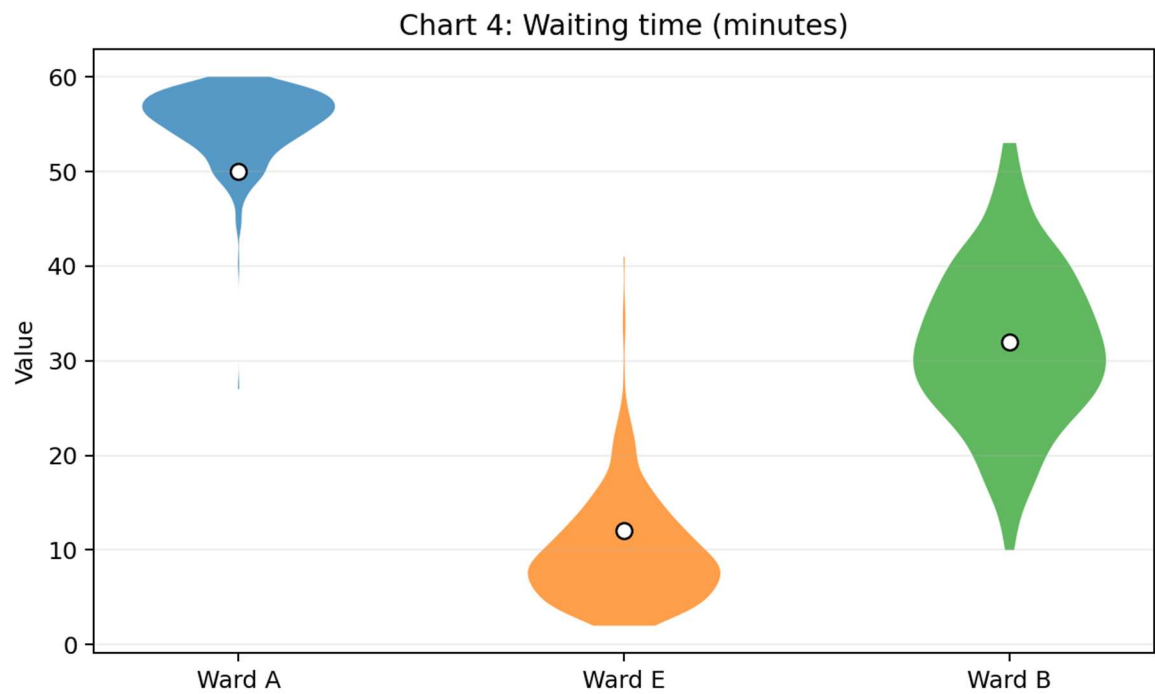


Chart 4: Waiting time (minutes)

Violin	Skew (answer)	Mean vs Median (answer)
Ward A	Left skew	Mean < Median
Ward E	Right skew	Mean > Median
Ward B	Symmetric	Mean $\approx$ Median

Interpretation: Ward A: Left skew, so Mean < Median.; Ward E: Right skew, so Mean > Median.; Ward B: Symmetric, so Mean $\approx$ Median.



Part B (2 marks)

B1. In Chart 4, violin “Ward A” is not symmetric. Towards which side is its tail, and which is the better measure of a typical waiting time here: mean or median? Give one reason. [1]

Answer: Ward A has a tail toward the lower-value side (left-skewed). The median is the better measure of a typical waiting time because the long lower tail can pull the mean downward, making the mean less representative of the central waiting time.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Answer:

```
ggplot(df, aes(x = group, y = value)) + geom_violin() + stat_summary(fun = median, geom = "point", color = "white", size = 2.5)
```

This maps group to the x-axis and value to the y-axis, draws one violin per group, and overlays the median as a point.

Final Answer Summary

Group	Skew	Mean vs Median
Brand Z	Right skew	Mean $>$ Median
Brand X	Left skew	Mean $<$ Median
Brand V	Left skew	Mean $<$ Median
Store 1	Symmetric	Mean $\approx$ Median
Store 4	Left skew	Mean $<$ Median
Store 2	Right skew	Mean $>$ Median
City	Left skew	Mean $<$ Median
Metro	Symmetric	Mean $\approx$ Median
Town	Right skew	Mean $>$ Median
Ward A	Left skew	Mean $<$ Median
Ward E	Right skew	Mean $>$ Median
Ward B	Symmetric	Mean $\approx$ Median